ECON 5 / POLI 5D

Introduction to Social Data Analytics

UC San Diego Summer 2026

Instructor: Torsha Chakravorty (tchakravorty@ucsd.edu)

- Lectures: Mondays & Wednesdays in Malk B90
- 11:00 AM to 12:50 PM

TA: Artur Bayramyan (abayramy@ucsd.edu)

- Lab sections: Fridays in Malk B90
- 12:00 noon to 1:50 PM

Office hours:

- Torsha Chakravorty (Instructor):
  - Mondays & Wednesdays in Malk 335
  - 1:30 PM to 2:30 PM
- Artur Bayramyan (TA):
  - Fridays on zoom
    - Zoom link: `https://ucsd.zoom.us/j/3460720818`
    - TA office: Malk 339
  - 10:00 AM to 11:00 AM

Important Dates

- Midterm: Wednesday, July 15
  - 11:00 AM to 12:50 PM (regular class time)
  - Location TBD
- Final: Saturday, Aug 01
  - Location & time TBD
- Weekly labs are due Sundays at 11:59 PM
- Quizzes are due Friday at 11:59 PM

Overview and course objectives

This course has four main goals. The first goal is to introduce you to interesting and important social science questions. Each chapter will introduce a different application, often highlighting research from faculty at UCSD. We will cover a wide range of topics, including how colleges promote intergenerational mobility, what motivates people to vote, how we identify discrimination in labor markets, among others.

The second goal is to show you how data can be used to inform our discussion of these topics. We will be using a lot of different datasets in this course. Some come from experiments that have been run by researchers. Others come from administrative datasets collected by governments. For many of the datasets, we will only have time to scratch the surface of what is possible with this data.

The third goal is to give you the tools to perform data analysis. We will focus on three popular software programs: Excel, Stata, and R. While learning coding fundamentals in each of these programs, we will use data to shed light on big social science questions.

A fourth goal is to help you learn how to think through a data problem before writing code. In class (and in exams), we will often practice writing pseudo-code: plain-language steps that describe what we want the computer to do before we translate those steps into Excel, Stata, or R syntax. This is an especially important skill now that AI tools can often help with the syntax of a particular coding language. The harder and more valuable skill is knowing what you want the code to do, why each step is needed, and how to check whether the output makes sense.

A final, related goal is to help you practice communicating what you are learning. Data analysis is not only about producing a number or a graph. It is also about explaining what you did, what you found, and what the limitations are. For this reason, class meetings will include some low-stakes opportunities to discuss your goals, your progress, your questions, and your interpretation of results.

Prerequisites

There are no prerequisites for this course. In particular, we do not expect any prior coding experience. There will be some math involved in interpreting linear equations.

Lectures and Labs

There will be two lectures per week. Each lecture will be 1 hour and 50 minutes long. Lectures are an important part of the course. Therefore, assessments and activities will take place in-person during class. Recordings, when available, are meant to help you review material; they are not a substitute for in-class participation.

In addition, all the lecture material, as well as bonus material, will be available on Canvas. In certain cases, I will ask you to watch a video before arriving at class. Make sure you watch this video so that you are prepared for the material in lecture. Additionally, there will be reading material that accompanies each lecture.

In addition to lecture, each week there will be a lab. Each lab will involve completing an Excel workbook, Stata Do-file, or R script. If you attend in person, you will work on sections of the lab together with your classmates who attend in person. Labs will almost always be in-person with one or two exceptions.

Each week you will need to answer a short quiz related to the lab. If you attend in person, the TAs will help you through the assignment and go over the relevant code. If you cannot attend in person, you can still get credit for the lab by completing the lab and then completing the quiz related to that lab. You should feel free to ask questions about quizzes and labs at office hours.

There may also be occasional ungraded in-class questions, polls, or short diagnostic quizzes. These are meant to help you and me understand what is clear, what is confusing, and what we should spend more time on. They are not designed to punish mistakes.

Course Materials

All of the course materials will be made available through Canvas. This includes the software we will be using in the class, as well as a link to the online textbook for the course.

Assessment

Your grade will be based on a combination of:

- **Lab (10%):** Each week you will need to complete the online quiz associated with the lab. If you attend lab, the TAs will cover how to write the code for the lab. The answers will usually depend on executing the code and reporting a result. You will need to do this on your computer; the exact answers will not necessarily be covered in lab. If you do not attend, you can still get credit by completing the lab on your own time and completing the associated Canvas quiz. The last question of every lab quiz will be to submit the completed lab script. **Important instructions: If you do not submit a completed lab script, you will automatically receive a zero on the lab, regardless of your other answers. You may also receive a zero if your script has substantial issues. You only have one opportunity to submit lab quizzes, unlike the weekly quizzes which give you multiple attempts.**
- **Quizzes (10%):** These are separate from the lab quizzes you will turn in. There will be quizzes at the end of week 2 and week 4. Quizzes will be posted on Wednesday and must be completed by Friday at 11:59 PM. In general, they will be a mix of multiple choice and occasional short answer questions. These are open-note/open-textbook, but you should work independently on the quizzes. **Important instructions: You are given 3 attempts with the highest score kept. Quiz questions are randomized, so you will potentially get a different set of questions on each attempt, and different students may receive different versions of the quiz.**
- **In-Class Participation and Contribution (15%):** This part of the grade is based on active participation in the course. This includes attendance, asking questions, answering questions, making suggestions, participating in group work, explaining your reasoning in simple pseudo-code exercises, and offering constructive comments about what is helping you learn. At different points in the quarter, I may invite students to briefly share their goals, their progress, or what they are finding difficult. These short presentations and check-ins are low-stakes and are meant to help you practice communicating about data analysis. Quality matters more than quantity: the goal is thoughtful engagement, not speaking just to speak.
- **Midterm (25%):** The midterm will be an in-person assessment held during the course. The exact date and time will be announced on Canvas. The midterm may ask you to interpret output, explain code, write short pieces of code, and write pseudo-code. For pseudo-code

questions, the goal is not to test whether you remember every piece of syntax perfectly. The goal is to assess whether you understand the sequence of steps required to solve a data problem. You need to be available to take the midterm on the scheduled midterm date. Exceptions are made only for verifiable medical, legal, university-related, or other approved reasons.

- **Final Exam (40%):** The final will be an in-person cumulative exam, but it will be weighted more heavily toward the R portion of the class. You need to be available to take the final on the scheduled final date. Exceptions are made only for verifiable medical, legal, university-related, or other approved reasons.

In several parts of the course, including in-class exercises and exams, you may be asked to write pseudo-code. Pseudo-code does not need to follow the exact syntax of Excel, Stata, or R. Instead, it should clearly describe the logical steps needed to solve a data problem. For example, you might be asked to explain how you would clean a dataset, construct a variable, merge two datasets, produce a graph, or estimate a regression before writing the exact code.

The course is generally graded on a relative curve. In past classes, generally you can be confident you will receive an A of some type (A+, A, or A-) if you are in the top 30 percent. You can be reasonably confident you will receive at least a B of some type (B+, B, or B-) as long as you are in the top 70–75 percent of the class. This is only a rough guide. These numbers are only meant to help you understand how well you are doing in the class. They are not concrete rules. In particular, there are objective standards as well. For example, if you get above a 90 percent as your final course grade, you will be guaranteed an A- at the very least.

Use of AI Tools

AI tools are now part of the world in which we work, and they can be useful. I encourage you to view AI a bit like an automobile: it can help you get somewhere faster, but sometimes the point of the activity is not simply to arrive. The point might be to build the muscle yourself. If your goal is to get fitter, you may intentionally choose to walk rather than drive.

That is how we will approach AI in this course. You may use AI tools to help you study, review concepts, generate additional practice questions, understand error messages, or ask for explanations of code. You should not use AI tools to complete graded work in a way that substitutes for your own understanding. In particular, you may not submit AI-generated answers as your own work, use AI to complete quizzes or exams, or use AI to produce code that you cannot explain.

This is also why we will emphasize pseudo-code. AI can often help you remember whether a command in R or Stata needs a comma, a parenthesis, or a particular option. But AI is much less useful if you do not know what you are trying to do. In this course, I want you to practice the part of coding that comes before syntax: identifying the task, breaking it into steps, deciding what variables are needed, and thinking about how to verify the result.

A good rule of thumb is this: if I ask you to explain what your code does (which I will do for graded exams), why it works, and how to interpret the output, you should be able to answer without relying on the AI tool. When in doubt, ask your TA before using AI on an assignment.

Academic Honesty and Plagiarism

All graded work must be done by you. If you are unfamiliar with the University's policy on academic integrity, please see `http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2`. There is a zero-tolerance policy for academic integrity violations, and if you are found to have violated the University's academic integrity policy you will receive a failing grade in the course.

Course FAQs

1. **What do I need to buy for the course?** We will use an online textbook that was written for the course and is available through Canvas for free. The software we will be using – Excel, Stata, and R – is also available through Canvas. Check the first module, which contains general course materials. One of the pages goes through all the installation instructions.
2. **Are there any technology requirements?** We will be using Excel, Stata, and R throughout the course. You can download these on either PCs or Macs, but Chromebooks are very difficult to download the programs on. There are computers in the library that have these programs installed. The Data and GIS Lab: `https://library.ucsd.edu/computing-and-technology/data-and-gis-lab/index.html` has computers that have these programs. However, if you attend lab in person, most of the lab will use your laptop to complete a coding assignment. If you do not have a laptop that is capable of downloading R and Stata, there is also a possibility for a long-term loaner laptop from the University. Please go here: `https://library.ucsd.edu/computing-and-technology/computing/index.html` to receive more information.
3. **How can I stay organized in this class?** In the Modules tab in Canvas, there will be a page named Week X: Plan for the Week, for every week in the course. This page includes a calendar of everything that is going on in the week, including readings, lecture slides, lecture code, and links to quizzes and labs. The final step for each week is a list of deliverables you are expected to finish by the end of the week.
4. **I have a question about the course. Where should I go to ask this question?**
   a. First, check the course website and the syllabus. For example, if the question is: is there a quiz this week, you should navigate the Modules and find the given week for the course. Many questions can be answered by looking through the Plan for the Week on Canvas.
   b. If you still cannot find the answer to your question, you can ask it on Piazza, which is linked from within Canvas. This should be where you post most of your questions. Please, however, do not post questions that include code or partial answers to quizzes or labs.
   c. If you have a question related to the course that you do not think is relevant for other students, you can send an email to your TA, or drop by during office hours.
5. **I'm on the waitlist. What are my chances of getting off?** Waitlists at UCSD are automated, and I cannot manually allow anyone off the waitlist. While we try to expand the class when there is excess demand, we are sometimes constrained by classroom size and TA availability.

6. **I think a question on the midterm or final was graded incorrectly. What should I do?** For midterms and finals, regrade requests will be done through Gradescope. For a regrade request, please state clearly why you think your solution was graded incorrectly.

Course Schedule

The schedule below lays out what is covered each week both in terms of the empirical application as well as the coding and software.

Week 1

(1) Introduction to Excel

- Empirical Application: Instructor Incentives and Student Performance, by Andy Brownback and Sally Sadoff (2020)
- Data tables
- Functions
- Pivot tables

(2) Introduction to Stata

- Empirical Application: Intergenerational Mobility Rates by College. Data comes from Opportunity Insights
- The Stata Graphical User Interface (GUI)
- Do-files
- Basic data analysis commands
- Interpreting and constructing histograms

Week 2

(3) Data Wrangling in Stata

- Empirical Application: Racial Discrimination in Traffic Stops. Data comes from the Stanford Open Policing Project
- Introduce concept of data wrangling
- Learn the append, merge, and collapse commands
- Bar charts in Stata
- Ways to improve data visualization in Stata

(4) Regression in Stata

- Empirical Application: Disrupting Education Using Technology, by Muralidharan, Singh, and Ganimian (2019)
- Estimate and interpret linear regressions in Stata

- Introduce concept of fitted values and residuals
- Visualize and plot the results of regressions in Stata

Week 3

(5) Introduction to R

- Empirical Application: Resume Experiments, by Bertrand and Mullainathan (2004)
- Objects and variables in R
- Introduction to data frames
- Subsetting data frames in R

(6) Midterm on Wednesday

Week 4

(7) Data Wrangling in R

- Empirical Application: The Rug Rat Race, by Garey Ramey and Valerie Ramey
- If statements
- For loops
- Introduction to the tidyverse package

(8) Data Visualization in R

- Empirical Application: China's War on Air Pollution, by Greenstone, He, Jia, and Liu
- Histograms, scatter plots, and box plots in R
- Using dates in R
- ggplot2

Week 5

(9) Linear Regression in R

- Empirical Application: The Butterfly Ballot
- Linear regression in R
- Plotting the regression line
- Predicted values and residuals

(10) Functions in R

- Empirical Application: The Impact of Unconditional Cash Transfers, by Haushofer and Shapiro
- Building your own functions in R